

At-Grade Highway–Rail Crossing Safety and Network Dataset (Jefferson County, Alabama)

This dataset was prepared to support future transport safety analysis and accessibility studies related to at-grade highway-rail crossings in Jefferson County, Alabama. All relevant datasets displayed in the interactive dashboard are available for download in various formats. Links to the raw datasets are provided, and the processed analytical summaries are attached.

Link to the interactive dashboard: <http://avinashu.com/>

Data sources -

1. Highway-Rail Grade Crossing Incident Data (Form 57)

- Coverage: Nationwide (1975–latest available year)
- URL: https://data.transportation.gov/Railroads/Highway-Rail-Grade-Crossing-Incident-Data-Form-57-/7wn6-i5b9/about_data

2. Highway-Rail Crossing Inventory Dataset

- Coverage: Nationwide crossing inventory
- URL: <https://data.transportation.gov/stories/s/Crossing-Inventory-Listing/ejv6-cpdh/>

Workflow and Data

From the Highway–Rail Grade Crossing Incident dataset, all records corresponding to Jefferson County, Alabama, were filtered. Incidents occurring since 2015 were extracted, and the data were aggregated annually.

The dataset was then grouped by Grade Crossing ID, reporting the total number of incidents per crossing per year. In parallel, incidents and fatalities were aggregated by both crossing and year, producing annual counts of crashes and total fatalities for each crossing.

These annual summaries were merged across the available inventory records, resulting in a consolidated dataset covering the last 10 years of incidents and fatalities at each grade crossing.

Finally, the incident data were joined with the geographic coordinates of the relevant crossings, and the resulting dataset was exported in both GeoJSON (for spatial analysis and mapping) and Excel formats for tabular analysis and reporting.

Data Dictionary of the processed analytical summary of the At-Grade Railroad Crossings Incidents in Jefferson County, Alabama

Column Name	Description
Crossing ID	Unique identifier for each at-grade highway–rail crossing.
Street	Name of the roadway intersecting the railway at the crossing.
Trains per day	Average number of trains passing through the crossing per day.
Total Incidents (Since 1975)	Total number of reported crossing incidents from 1975 to present.
Total Fatalities (Since 1975)	Cumulative fatalities resulting from crossing incidents since 1975.
Total Injuries (Since 1975)	Cumulative non-fatal injuries resulting from crossing incidents since 1975.
Total Incidents (2015–2025)	Number of reported crossing incidents between 2015 and 2025.
Total Fatalities (2015–2025)	Number of fatalities from crossing incidents between 2015 and 2025.
Total Injuries (2015–2025)	Number of non-fatal injuries from crossing incidents between 2015 and 2025.
latitude	Latitude of the crossing location (decimal degrees).
longitude	Longitude of the crossing location (decimal degrees).
Annual average daily traffic count	Average daily vehicle traffic at the crossing roadway.
Annual average daily traffic year	Year the Annual Average Daily Traffic (AADT) value was reported.

Basemap Layers

Layer	Purpose	Notes	Data Source
Road (Network dataset with nodes)	For representing local road networks not captured in FHWA Functional classification.	Includes network topology (nodes/edges) for analysis	Downloaded from Overpass Turbo - https://overpass-turbo.eu/
Road (Classified – functional hierarchy)	Show road classification in 8 classes as per FHWA classification	Classified according to FHWA Functional Classification (e.g., Interstate, Arterial, Collector, Local); used for hierarchy visualization	City of Birmingham GIS Mapping Files - https://data.birminghamal.gov/dataset/gis-mapping-files ; FHWA Functional Classification Guidance - https://www.fhwa.dot.gov/policyinformation/hpms/hfcccp.cfm
Rail Network – North American Rail Network (Lines)	Represents rail line infrastructure across North America for mapping and analysis	Comprehensive rail line dataset covering U.S., Canada, Mexico; includes ownership, type, passenger/freight info, STRACNET attributes	NTAD – Federal Railroad Administration (FRA) / Bureau of Transportation Statistics (BTS): https://catalog-old.data.gov/dataset/north-american-rail-network-lines3
County / Federal Boundaries	County boundaries, census tracts	County boundary for Jefferson County	TIGER/US Census - https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html